

A comparative neutronic analysis of UN-ThN vs. traditional UO₂ fuel in a quasi-infinite pin-cell lattice at beginning of life

- **Author:** Clément DAROUSSIN
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ABSTRACT

This study presents a preliminary neutronic assessment of Uranium-Thorium Nitride (UN-ThN enriched with N-15) fuel as a high-performance alternative to traditional Uranium Dioxide (UO₂) in pressurized water reactors [1]. Using the OpenMC Monte Carlo code and the JEFF-3.3 continuous energy library [4], a quasi-infinite pin-cell model (utilizing reflective boundary conditions radially, while accounting for axial leakage through 20 cm water reflectors modeled at the top and bottom of the active fuel length [2]) was developed to evaluate Beginning-of-Life (BOL) characteristics. The comparative analysis focuses on three primary metrics: inherent safety, neutronic spectral evolution and the breeding potential. Results indicate a 9.47% reactivity penalty for the UN-ThN fuel, counterbalanced by a 39.97% improvement in the conversion ratio and a 72.21% increase in the magnitude of the negative Doppler feedback [5] coefficient. These findings suggest that while the nitride-thorium matrix requires specific isotopic enrichment (N-15) and higher initial fissile loading, it offers a superior pathway toward enhanced passive safety and sustainable fuel cycle management.

1. Introduction

The evolution of Small Modular Reactors (SMRs), Generation IV systems and modern large-scale Pressurized Water Reactors (PWRs), notably the EPR2 fleet, necessitates the exploration of advanced fuel matrices capable of providing enhanced inherent safety and improved resource sustainability within existing technological frameworks, driven above all by the world's rapidly growing energy demands. The Uranium Dioxide (UO₂) remains the industrial standard and its established fabrication infrastructure and predictable performance, its neutronic and thermal limitations under high burnup or extended cycle constraints are well-documented. That's why it will serve as the reference fuel.

This R&D project investigates the potential of a Uranium-Thorium Nitride (UN-ThN) fuel matrix (80% UN – 20% ThN) vs. the standard UO₂ both enriched at 4.95% [1, 7]. Nitride fuels are distinguished by their higher heavy-metal density and superior thermal conductivity compared to oxide counterparts, providing a more robust safety margin against thermal-induced power excursions. There are two particular interests of this study. First, the UN-ThN matrix analyzed here is enriched in N-15 to counter the N-14 parasitic absorption reaction (without considering the associated industrial economic constraints). Furthermore, unlike most contemporary

research that leverages nitride density for reactor compaction (higher heavy-metal density) in Small Modular Reactors (SMRs), this work explicitly evaluates the UN-ThN matrix within the framework of large-scale commercial reactors, such as the EPR2 [6]. The objective of this Phase 01 study is to establish a rigorous neutronic baseline at the Beginning of Life (BOL) using the OpenMC 15.3.0 transport code. This research focuses on the core neutronic trade-offs, quantifying the relationships between initial reactivity, spectral hardening, inherent safety performance, and breeding potential. Following this initial assessment, future phases will address the fuel's depletion behavior to evaluate its performance over a full reactor operational cycle.

2. Computational framework and simulation parameters

This section details the computational framework adopted for this phase. First, Table 1 outlines the core design and geometrical parameters of the unit pin-cell, based on a standard PWR geometry with axial leakage. Next, Table 2 summarizes the fundamental physics and nuclear data, including the evaluation libraries and thermal-hydraulic states. Finally, the specific Monte Carlo simulation parameters, tracking statistics, and numerical convergence criteria are compiled in Table 3.

Parameter	Value	Details
Cell Type	Unit pin-cell (3D)	Axial leakage added
Fuel Radius (r1)	0.4096 cm	Standard PWR pellet
Clad Outer Radius (r3)	0.4750 cm	Zircaloy-4 cladding
Pitch	1.26 cm	Lattice spacing (X, Y axes)
Active Fuel Height	420.00 cm	EPR2 standard active length
Axial Reflectors	20.00 cm (x2)	Light water
Total Model Height	460.00 cm	Cumulative Z-axis length
Boundary Conditions	Reflective / Vacuum	Reflective on X/Y (quasi-infinite lattice)

Table. 1 - OpenMC configuration - design & geometry [2]

Parameter	Value	Status
Nuclear Library	JEFF-3.3	HDF5 format
Energy Treatment	Continuous	High-fidelity resolution
Fuel Temperature	900K (then 1200K)	//
Other Temperature	600K	//
Thermal Scattering	c_H_in_H2O	S(a, b) applied
Transport Mode	K-Eigenvalue	Neutron transport only

Table. 2 - Physics & nuclear datas

Parameter	Value	Detail / Impact
Transport Mode	Monte Carlo	K-eigenvalue (OpenMC)
Particles per batches	50,000	//
Active Batches	250	Variance reduction
Inactive Batches	50	Source stabilization
Total Histories (N)	12,500,000	$\sigma \approx 30$ pcm (0.03%)

Table. 3 - Monte Carlo simulation parameters and convergence criteria

3. Materials definition

This section defines the isotopic compositions and physical densities of the materials within the pin-cell. The composition parameters used in this simulation are derived from the literature [1].

Fuel material	Density (g/cm ³)	U-235 enrichment (wt%)	Heavy metal mass (×10 ³ kg)
UO ₂ (Reference)	10.422	4.95	16.21
UN-ThN (Test)	10.792	4.95	17.37

Table. 4 - Isotopic composition and physical density of fuel matrices [1]

The reference UO₂ fuel is defined with a physical density of 10.422 g/cm³ and a fixed enrichment of 4.95 wt% U-235 [2]. This specific value is the practical limit for most existing commercial fabrication facilities (often referred to as "low enriched uranium" or LEU). It represents the maximum reactivity potential available in current Generation III PWRs to optimize fuel cycle length. Using this value as a reference for both geometry and materials ensures a consistent baseline that strictly follows the parameters defined in the reference study [1]. Furthermore, this strategy aligns the benchmark with the

operational standards of the EPR2, which serves as the industrial model for this simulation [6]. This "high enrichment" approach also maximizes the neutronic contrast between the fuels, making it easier to differentiate and analyze the specific performance benefits of the U-Th matrix within a next-generation reactor environment.

The uranium-thorium nitride (UN-ThN) mixture (80% UN + 20% ThN) is an advanced fuel candidate currently being studied for Generation IV reactors and specialized SMRs. Nitride fuels are investigated as alternatives to oxides due to their higher heavy-metal density and superior thermal conductivity [7]. The use of natural Nitrogen would introduce a significant reactivity penalty due to the $^{14}\text{N}(n,p)^{14}\text{C}$ reaction, which acts as a strong thermal neutron poison. To ensure optimal neutronic economy, the nitride composition typically assumes enrichment in N-15, as natural nitrogen (N-14) acts as a strong neutron poison through the (n,p) reaction [1]. However, the economic feasibility of this isotopic enrichment remains a key aspect that will require further investigation, particularly regarding its impact on the overall fuel cycle cost. This benchmark adopts industry-standard structural materials to ensure results are applicable to broader PWR technology. The cladding is modeled as Zircaloy-4, a standard nuclear-grade alloy, the gap is filled with Helium and the coolant/moderator is conventional light water [2]. This configuration provides a more conventional environment to evaluate the transition from UO₂ to UN-ThN.

4. Geometry Setup

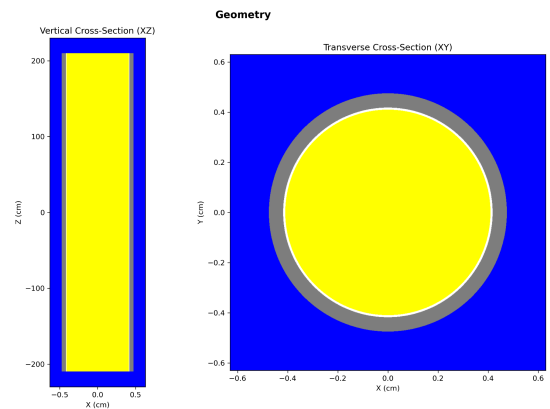


Figure. 1 - Unit pin-cell geometry [2] (fuel in yellow, clad in grey, gap gas in white and moderator in blue)

The geometry is modeled as a quasi-infinite pin-cell lattice using reflective boundary conditions and axial leakage. The dimensions are taken from the VERA Core Physics Benchmark specifications [2], which are

based on the industrial standard of Pressurized Water Reactors (PWR) such as the EPR2.

5. Simulation configuration

The eigenvalue mode is used to determine the system's k-eff and a uniform spatial source is defined within the fuel region with the use of 'only_fissionable = True' to ensure that the initial neutron births are strictly confined to the fuel pellet. To achieve a satisfying statistical precision, the number of active particles was set at 10,000,000 distributed over 200 batches (50,000 each batch). For a simple pin-cell geometry [2], the Shannon Entropy typically stabilizes around the 20th cycle, a conservative approach was adopted, using 50 inactive batches to ensure full source convergence. The neutron flux spectrum is resolved using 5,000 energy groups, while the radial power distribution inside the pellet is calculated using a 'CylindricalMesh' with 250 radial divisions. To determine the conversion ratio, radiative capture ((n,gamma)) and fission reactions are tracked alongside the total absorption of Th-232, U-238, and U-235.

6. Criticality

Fuel	k-eff	Std Dev
UO ₂	1.376	30.5 pcm
UN-ThN	1.245	27.8 pcm

Table. 5 - Comparison of multiplication factors (k-eff)

The transition from UO₂ to UN-ThN results in a significant reactivity penalty of approximately 9.47%. The consistency in the standard deviation (around 0.03%) confirms that the source convergence (Shannon Entropy) was successfully reached before the active batches began. Even if from a design perspective, this value seems to quantify the additional enrichment required to achieve the same cycle length as standard UO₂, we will show with a CR calculation that this is a 'controlled investment' : the neutrons 'lost' for reactivity are actually intercepted by Th-232 to breed new fissile material [5]. In the next study, a depletion analysis will show that the UN-ThN reactivity curve remains significantly flatter as burnup increases.

7. Safety (Doppler Coefficient)

The Doppler Coefficient (α_D) is a safety metric used to quantify the reactor's ability to self-stabilize when temperature rises. As fuel temperature rises, the increased thermal motion of nuclei causes an effective broadening of absorption resonances (primarily in U-238 and Th-232), increasing the resonance absorption and removing neutrons from the chain reaction, which instantly lowers

the reactivity. Comparing α_D between UO₂ and UN-ThN is essential to determine if the transition to a denser nitride matrix or the introduction of Thorium alters the safety margin. Since Th-232 has a different resonance profile than U-238, we must verify that the UN-ThN fuel maintains a sufficiently negative feedback to prevent power excursions [5]. The Doppler coefficient is calculated by measuring the change in reactivity (ρ) resulting from a temperature increase, expressed as:

$$\alpha_D = \frac{\Delta\rho}{\Delta T} = \frac{(\rho(T_{hot}) - \rho(T_{ref}))}{(T_{hot} - T_{ref})} \quad (\text{in pcm/K})$$

Metric	UO ₂	UN-ThN	Comparison
Reactivity Swing ($\Delta\rho$)	-527.0 pcm	-907.6 pcm	+72.21%
Doppler Coeff (α_D)	-1.76 pcm/K	-3.03 pcm/K	Enhanced Stability

Table. 6 - Doppler Coefficient (α_D), sensitivity and feedback analysis

The results from the safety transient analysis confirm that both fuel candidates possess an inherent self-stabilizing mechanism, as indicated by the negative values of the Doppler Coefficient (α_D). The UN-ThN fuel exhibits a significantly stronger Doppler feedback (72.21% more negative than UO₂) meaning that it would lower the reactivity twice more than standard oxide fuel in the event of an accidental power excursion due to the higher atomic density of the nitride matrix combined with the specific resonance capture integrals of Th-232 [5].

8. Neutron spectrum

The following figure shows the neutron spectrum per unit lethargy ($\Phi(u)$) (to prevent spectrum distortion caused by varying energy bin widths). Identifying the thermal peak validates the thermalization process and reveals potential spectrum hardening depending on the fuel composition.

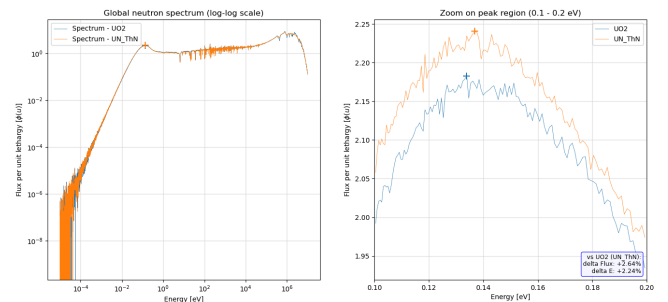


Figure. 2 - Neutron flux spectrum per unit lethargy

Parameter	UO ₂	UN-ThN	Relative Difference
Peak Energy	0.1339 eV	0.1369 eV	+2.24%
Magnitude	2.1828	2.2405	+2.64%

Table. 7 - Characterization of neutron spectrum peaks

The high atomic density of the Nitride matrix (UN-ThN) increases the macroscopic absorption cross-section for the lowest-energy thermal neutrons. This preferentially removes "colder" neutrons, shifting the thermal equilibrium peak to the right (+2.24% in agreement with [1] where the UO₂ and UN-ThN peaks are found to be very close.), it confirms a slightly harder spectrum while remaining within the thermal regime. The 2.64% increase in the thermal peak magnitude indicates a higher concentration of the thermal neutron population within the lattice. This is primarily attributed to the optimized neutron economy achieved through the use of N15 enrichment. By eliminating the parasitic (n,p) absorption reaction associated with natural Nitrogen, the nitride matrix allows a greater proportion of the neutron population to successfully thermalize in the moderator and remain available for fission. This 'spectral concentration' effect compensates for the marginal spectral hardening caused by the higher heavy-metal density, ensuring that the thermal neutron economy is not only preserved but slightly enhanced compared to the UO₂ reference [1].

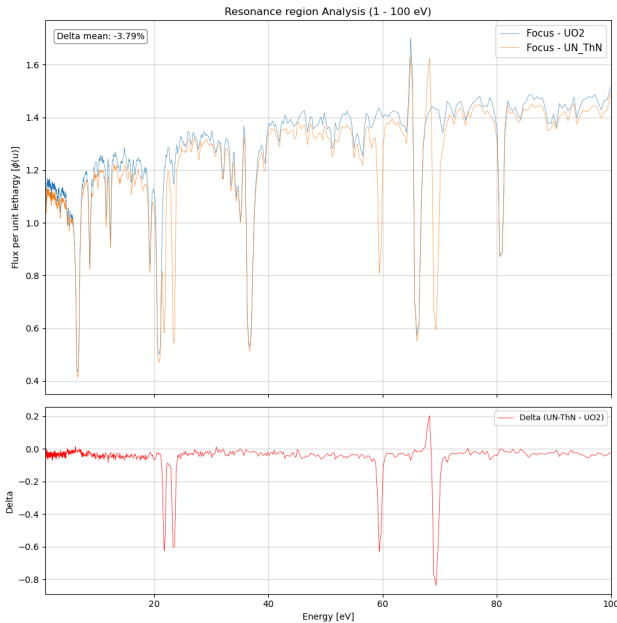


Figure. 3 - Focus on the epithermal resonance region (1–100 eV)

In the resonance region (1–100 eV), the integrated relative flux difference is -3.79%. This value captures the global epithermal flux depression, encompassing both the persistent background difference (ranging between -3% and -4% due to the increased heavy metal loading) and the localized flux dips. The most significant depressions are observed at 9.5 eV, 21.8 eV, 23.5 eV, and 69.2 eV, which correspond precisely to the major capture resonances of Th-232 :

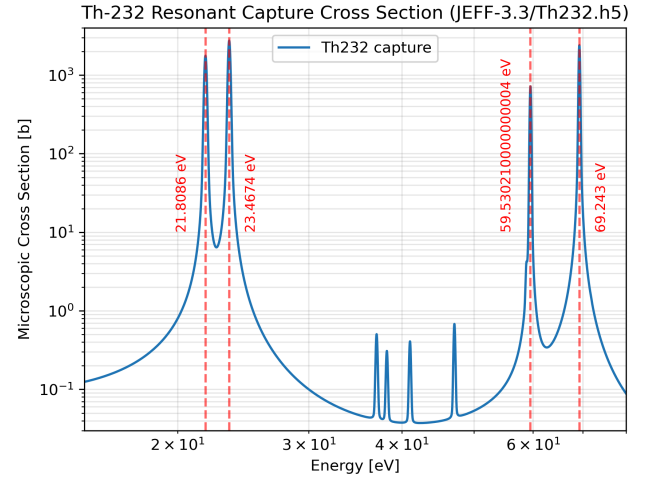
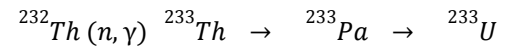


Figure. 4 - Direct plot of the Jeff-3.3's cross section of Th232 using the native `"openmc.plot_xs()"` method

It confirms that the UN-ThN fuel acts as an efficient fertile sink. These neutrons are not simply lost but are captured to initiate the transmutation chain into fissile U-233 :



9. Radial power distribution

9.1 Shape comparison

The radial power distribution (in [fissions/(cm³ · src-n)]) allows us to characterize the spatial self-shielding effect and compare how high heavy-metal density and Thorium's cross-sections influence fission homogeneity. The following 2D heatmap (employing a shared scale) allows us to quickly visualize the fission density across the entire pellet cross-section.

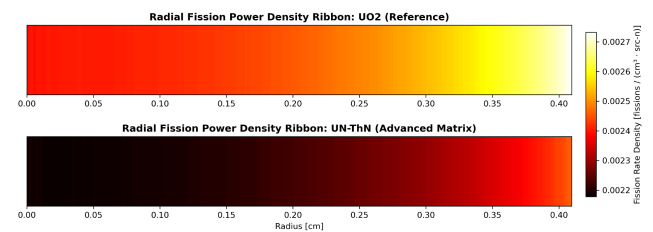


Figure. 5 - 2D heatmap of fission density distribution

To obtain actionable results, a quantitative radial profile is required :

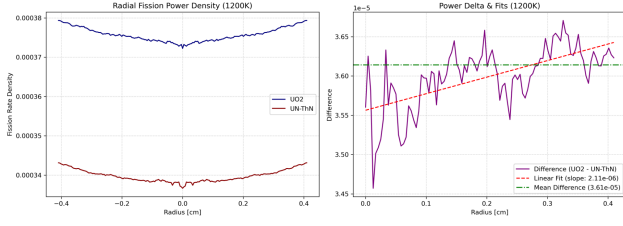


Figure. 6 - UO₂ vs. UN-ThN radial power profile at 1200 K

Analyzing the slope of the delta curve indicates how the profile shape evolves, while the mean of the deltas quantifies the global power shift. The profile factor (PF = P_{max}/P_{avg}) quantifies the local repartition (close to 1 means that the profile is quasi-flat), while the shielding index (e^{slope}) quantifies the spatial self-shielding effect (an indispensable metric for evaluating the fuel's thermal performance and safety margins).

Fuel	Slope	Shielding index (e^{slope})	Δ Index	Curve of difference's slope	Mean of the difference (std dev)
UO ₂	0.0382	1.0389	0.0022	2.107e-06	3.61e-05 (2.8764e-07)
UN-ThN	0.0360	1.0367	/	/	/

Table. 8 - UO₂ vs. UN-ThN radial power profile and spatial self-shielding indices comparison

The globally positive mean of deviations (3.61e-05) confirms that the UO₂ profile lies slightly above that of the UN-ThN. Combined with an extremely low standard deviation (2.8764e-07), it reveals a near-perfect translation between the two curves with a low structural deformation. To understand the internal dynamics of the pellet, it is necessary to analyze the spatial self-shielding (a process by which thermal neutrons are preferentially captured at the periphery). The self-shielding index (e^{slope}) of UO₂ fuel is 1.0389, whereas the UN-ThN is 1.0367 indicating that both fuels are in a low self-shielding regime. The 0.0022 difference reflects a physically coherent trend but is too small to constitute an operationally significant difference between UO₂ and UN-ThN. It nevertheless reveals that the UN-ThN power profile is flatter, indicating that neutrons penetrate more easily to the center of the fuel. It is confirmed by the analysis of the difference curve. The positive slope (2.107 x 10⁻⁶) reveals that the UO₂ profile rises faster and reaches a higher level at the periphery, therefore the UN-ThN profile is flatter. Consequently, spatial self-shielding is more pronounced in UO₂.

Finally, neither UO₂ nor UN-ThN suffers from a severe shielding effect and the offset (3.61×10⁻⁵) with a standard deviation of 2.87×10⁻⁷ shows that the two profiles

are quasi-superimposable. UN-ThN is slightly superior, presenting less radial thermal gradient, hence lower mechanical stress on the pellet under transient conditions, a more homogeneous fission distribution across the pellet radius, a more radially uniform burnup and therefore a better utilization of the fissile atoms present at the center. However, this advantage remains modest at BOL. The gain in flux flatness offered by UN-ThN will only translate into a tangible operational benefit at high burnup, where the differences in peripheral neutron behavior between the two fuel types will become more pronounced.

9.2 Thermal sensitivity (900K vs. 1200K)

To characterize the temperature dependence of power profiles, we analyze the evolution of the shielding index for each fuel, as resonance broadening with rising temperature directly impacts spatial self-shielding :

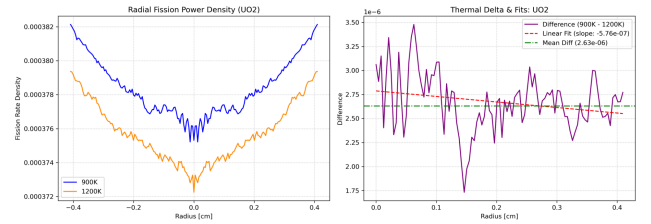


Figure. 7 - UO₂ radial power profile variation with T°

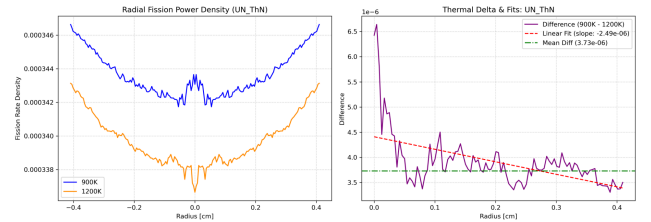


Figure. 8 - UN-ThN radial power profile variation with T°

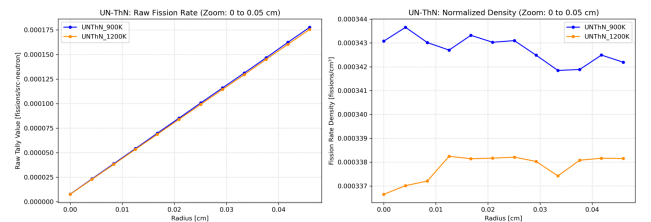


Figure. 9 - Zoom on the central region of UN-ThN profile

A local anomaly ($r=0$) in the UN-ThN profile shows a peak at 900 K inverting into a depression at 1200 K, accompanied by vertically symmetrical secondary oscillations. Two hypotheses were confronted: a numerical artifact due to the geometric singularity in cylindrical coordinates (the volume of the first mesh element being

three times smaller than the second, artificially amplifying the fission density) or a real physical effect tied to the transparency of N-15 combined with the Doppler broadening of Th-232. The plot of the unnormalized raw fission tally definitively settles the question: the curves at both temperatures appear strictly monotonic and increasing. The anomaly and its secondary oscillations therefore stem solely from a volumetric normalization instability.

Fuel	Temp	Slope	eSlope	Δ Index	Curve of difference's slope
UO ₂	900K	0.0364	1.0371	-0.0019	-5.7555e-07
UO ₂	1200K	0.0382	1.0389	-	-
UN-ThN	900K	0.0284	1.0288	-0.0079	-2.4896e-06
UN-ThN	1200K	0.0360	1.0367	-	-

Table. 9 - Radial power profile 's sensitivity to temperature variation

The UN-ThN thermal sensitivity is more pronounced than the reference fuel UO₂. Its slope of the difference curve is nearly four times more important than the UO₂ increment and it is the same about the shielding index variation. This sharper evolution reflects a stronger sensitivity of this fuel to temperature about the power distribution homogeneity (due to the stronger coupling between temperature and resonance capture of the thorium). Despite this steeper evolution, UN-ThN at 1200 K remains better to UO₂ in terms of absolute self-shielding level (e^slope of 1.0367 vs. 1.0389), meaning it does not significantly surpass UO₂ in shielding intensity at operating temperature.

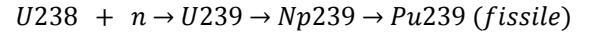
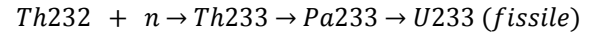
In conclusion, comparing the thermal sensitivity between 900 K and 1200 K identifies UN-ThN as the more advantageous choice for reactor safety and efficiency. Despite its more pronounced Doppler reactivity, the absolute self-shielding level at 1200 K (1.0367) remains lower than that of UO₂ (1.0389). This flattened profile drastically reduces the radial thermal gradient within the fuel rod. It minimizes local mechanical stresses during power transients (such as pellet-cladding interaction (PCI) and pellet cracking) which, combined with the inherently high thermal conductivity of nitride fuels, significantly lowers centerline temperatures. This broadens the margin to fuel melting, thereby offering significantly higher thermomechanical safety margins than the reference UO₂.

10. Conversion ratio

The conversion ratio (CR) is defined as follows:

$$CR = \frac{N_{\text{fissile nuclei produced}}}{N_{\text{fissile nuclei disappeared}}}$$

Production (numerator) is the total sum of radiative capture (n, γ) events involving all fertile isotopes present in the fuel (here: Th-232 and U-238). Each neutron captured here represents the birth of a future fissile nucleus through a chain of beta- decays:



Consumption (denominator) is the total sum of disappearance of fissile nuclei U-235, as well as any U-233 or Pu-239 produced over time, occurring via fission or sterile absorption.

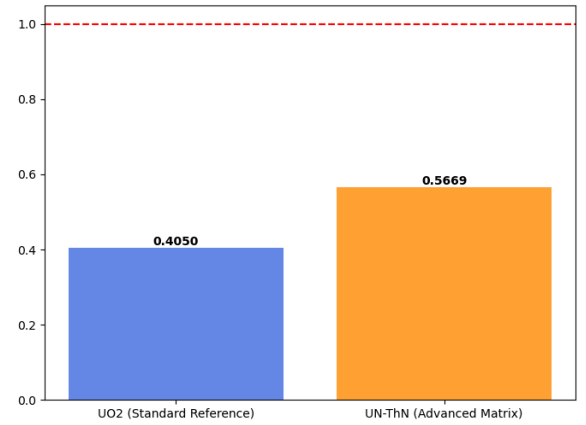


Figure. 10 - Comparison of conversion ratio (CR)

Both fuels remain below the breeding threshold (CR < 1), meaning they act as net fissile burners, rather than breeders. However, the UN-ThN fuel exhibits a higher conversion ratio (+39.97% extremely close to the 40% difference found in the Ishraq & Roy (2024) article [1]), demonstrating a much higher efficiency in converting fertile nuclei into new fissile material (specifically U-233 from 231Th). As anticipated, the transition to UN-ThN induces a substantial reactivity penalty of approximately 9.47% is directly compensated in the long term by the enhanced conversion performance of the thorium-nitride matrix and supports the potential for significantly extended fuel cycles (we'll check that in the next comparison study with a full depletion cycle).

11. Conclusion

The transition from the oxide reference to the advanced nitride-thorium matrix introduces an initial

reactivity penalty of approximately 9.47% which is a highly controlled investment in neutrons economy. Rather than being lost to parasitic captures, these neutrons are efficiently intercepted by the prominent epithermal resonances of Th-232 (notably at 9.5 eV, 21.8 eV, 23.5 eV, and 69.2 eV), leading to a 39.97% improvement in the conversion ratio (CR) even at Beginning of Life (BOL). This fertile-to-fissile transmutation chain (Th-232 to U-233) strongly supports the potential for a significantly lower reactivity swing (k-eff evolution over burnup) and extended fuel cycle lengths, to be confirmed in subsequent depletion phases.

From the standpoint of reactor safety and efficiency, UN-ThN exhibits clear superiority over UO₂. Regarding inherent safety feedback, the UN-ThN matrix delivers a 72.21% increase in the magnitude of the negative Doppler feedback coefficient, ensuring an enhanced and prompt self-stabilizing mechanism (capable of mitigating power excursions). This safety advantage is further complemented by optimized power distribution and thermomechanical margins; despite showing a thermal sensitivity nearly four times more pronounced than UO₂, the absolute spatial self-shielding level of UN-ThN at 1200 K remains strictly lower than that of UO₂ [7].

This reduces the radial thermal gradient within the fuel rod, thereby minimizing local mechanical stresses during power transients and effectively mitigating the risks associated with pellet-cladding interaction (PCI) and pellet cracking. When combined with the higher thermal conductivity of nitride fuels, this significantly lowers centerline temperatures and broadens the margin to fuel melting [7]. Therefore, despite the industrial and economic requirements associated with N-15 enrichment and higher initial fissile loading, UN-ThN seems to stand out as a superior pathway toward enhanced passive safety and sustainable fuel cycle management in next-generation light water reactors [1].

12. limits of the study and next steps

This study presents however three main technical limitations. First, even if this neutronic assessment clearly gives the advantage to UN-ThN, making it appear as an ideal solution on paper, these results must be put into practical perspective. To fully adopt this fuel for widespread commercial use, the harsh chemical, technical, and economic realities of a real-world nuclear reactor must be thoroughly assessed. Indeed nitride fuels are known for reacting violently with high-temperature, pressurized water in the event of a cladding breach and causing structural degradation of the fuel pellet. The parasitic

production of radioactive Carbon-14 is also a major issue. The use of N-15 is mandatory however its enrichment at an industrial scale still remains economically restrictive. Finally, the current fuel cycle is incompatible : existing reprocessing plants are exclusively tailored for oxides, and dissolving nitrides would require entirely new processes. This does not mean that implementing UN-ThN is impossible or even a flawed concept, but it highlights that further investigations must be pursued beyond the scope of neutronics. Secondly, this phase is strictly limited to Beginning-of-Life (BOL) conditions, whereas the true operational and sustainability advantages of the thorium-nitride cycle can only manifest over time. Finally, the use of a quasi-infinite 3D pin-cell model inherently neglects radial leakage and neighborhood effects (such as flux sharing and spectral transitions between adjacent assemblies) [2], which would be present in a full-core heterogeneous configuration. To address some of these limitations, the next phase of this research will involve a comprehensive depletion analysis using OpenMC's depletion module. This future work will track the isotopic evolution of both matrices over a 441-days operational cycle, specifically modeling the in-situ transmutation of fertile Th-232 into fissile U-233. This depletion study aims to confirm that the high-converter nature of the UN-ThN matrix successfully mitigates its initial 9.47% reactivity penalty, ultimately achieving a flatter reactivity swing, a higher discharge burnup, and superior resource utilization compared to standard UO₂ [1].

References

- [1] [M. A. R. Ishraq et S. Roy. « A comparative neutronic analysis of U-Th fuels as potential alternatives to traditional UO₂ fuel for enhanced performance and safety in the ACP-100 SMR », *International Journal of Advanced Nuclear Reactor Design and Technology*, vol. 6, n° 2, p. 177-184, juin 2024.](#)
- [2] [A. T. Godfrey. "VERA Core Physics Benchmark Progression Problem Specifications," Oak Ridge National Laboratory \(ORNL\), Oak Ridge, TN \(United States\). Tech. Rep. CASL-U--2013-0281-000, Mar. 2013.](#)
- [3] [Reuss, P. Précis de neutronique. EDP Sciences.](#)
- [4] [JEFF Project. "LWR PIN CELL BENCHMARK INTERCOMPARISONS: An intercomparison study organised by the JEFF Project, with contributions from the UK, France, Germany, the Netherlands, Slovenia and the USA," OECD Nuclear Energy Agency \(NEA\)](#)
- [5] [W. J. M. de Kruijf. "Reactor physics analysis of the pin-cell Doppler effect in a thermal nuclear reactor." Ph.D. dissertation, Dept. Nuclear Engineering, Technische Univ. Delft, Petten, Netherlands, 1995. Report No. ECN-R-94-033.](#)
- [6] [C. Sandrin. « Modélisation neutronique du réflecteur pour le calcul des cœurs des réacteurs nucléaires à eau pressurisée : application à l'EPR », Thèse de doctorat, Physique, Université Paris 11, Sceaux, France, 2010.](#)
- [7] [B. Szpunar. "First principles investigation of the electronic-thermal transport of ThN, UN, and ThC." *Nuclear Materials and Energy*, vol. 32, p. 101212, Sep.](#)